

Parabolic multipliers cannot be Liouville numbers

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Conjecture 00000000377

Claim addressed. Both versions of the source claim Hausdorff dimension 1 for the parameters on the Mandelbrot boundary whose quadratic map has a parabolic periodic point with a Liouville multiplier. This set is empty.

For $c \in \mathbb{C}$, let $F_c(z) = z^2 + c$, and let $F_c^{\circ n}$ denote composition iteration. If $F_c^{\circ n}(z) = z$ with $n > 0$, its multiplier is

$$\lambda = (F_c^{\circ n})'(z).$$

A parabolic periodic multiplier is a root of unity: there is $q > 0$ with $\lambda^q = 1$. This is the usual definition, including the multiplier-1 convention as a special case; see Milnor [1]. It differs from an irrationally indifferent multiplier.

Proposition 1. *No parabolic periodic point of any quadratic map F_c has a multiplier that is a real Liouville number. Consequently the parameter set in the source has Hausdorff dimension 0, rather than 1.*

Proof. Suppose such a multiplier were the Liouville number $\xi \in \mathbb{R}$, viewed as a complex number. Then $\xi^q = 1$ for some positive integer q . Taking absolute values gives $|\xi|^q = 1$, hence $|\xi| = 1$ and $\xi \in \{1, -1\}$. Both possibilities are rational, whereas a Liouville number is irrational. This is a contradiction.

The irrationality fact can be seen directly from the standard Liouville approximation condition: for every positive integer N , there are integers a and $b > 1$ with

$$0 < \left| \xi - \frac{a}{b} \right| < \frac{1}{b^N}.$$

If $\xi = u/v$ were rational with $v > 0$, every distinct rational a/b would have distance at least $1/(vb)$. Taking N with $2^{N-1} \geq v$ contradicts the displayed inequalities, since $b \geq 2$. The Lean proof uses Mathlib's genuine Liouville condition and its proved irrationality theorem.

This obstruction holds for every parameter c , so it also holds after restricting to the actual Mandelbrot boundary. More explicitly, define

$$M = \{c \in \mathbb{C} : \{F_c^{\circ n}(0) : n \geq 0\} \text{ is bounded}\}.$$

The set of $c \in \partial M$ admitting the stated periodic point and multiplier is therefore empty. The Hausdorff dimension of the empty set is 0 under the usual nonnegative convention, as in Mathlib, and in particular is not 1. $\square$

Formal correspondence. Lean defines actual quadratic composition iterates and proves their evaluation agrees with function iteration. The multiplier is the actual complex derivative, with a proved polynomial-derivative bridge. The parameter set uses the actual bounded-critical-orbit Mandelbrot set and its frontier, periodicity, the parabolic root-of-unity condition, and the genuine Liouville property. It proves emptiness and actual $\dim_{\mathbb{H}}$ zero. Emptiness also holds without a Julia-set restriction, so imposing that restriction cannot add points. Only the first dimension claim is addressed.

References

- [1] J. Milnor, *Periodic orbits, external rays and the Mandelbrot set: an expository account*, Astérisque **261** (2000), 277–333. Original article.